

Geometry and measured range-law behavior. Panel a shows 8064 non-BOS token projections in the frozen (v1,v2) basis at the same layer-27 down-input site as Figure 1. Token coordinates are standardized separately; unit free-direction projections are overlaid with a common multiplier of 1.0 and measured in-plane lengths 0.0150 (full Hadamard) and 1.0000 (PrismQuant). Direction cosines and token standard deviations are distinct quantities. Percentile-initialized bounds expand to all observed points, avoiding clipping. Panel b preserves cumulative uncentered energy curves for layers 1, 13, and 27 over $k=1-256$ on a logarithmic x axis. Panel c is supplied as three matched standalone views: 2520 E1c activation observations; 280 E7 V-cache plus 112 E20 multi-slot observations; and 5342 E26 per-expert observations on Qwen3-30B-A3B-Base, one per routed expert, with the expert rather than the layer as the unit and f measured at its own rank $k \leq 6$. All three show range relative to the paired Hadamard result against $\sqrt{1-f}$, the identity line, and the same pooled linear fit over the 2912 points of the three Llama-3.2-3B reference sources ($R^2=0.861389$). The fit is not estimated separately for any view; the MoE observations are held out of it entirely and their own fit ($R^2=0.4328$) is stated in that panel. Insets enlarge the 0.85–1.0 corner without omitting any point from the main panels.